Week 9 (16.03.2026)

Auctions and mechanism design

① A seller runs a second-price, sealed-bid auction to sell a single object.

(a) there are two bidders who have independent, private values v_i $(i = 1, 2)$, which are either 1 or 2. For each bidder, the probabilities of values 1 and 2 are both 1/2. If there is a tie at a bid of x for the highest bid, then the winner is selected at random from among the highest bidders and the price is x.
How much can the seller expect to receive for their object?

There are 4 possible pairs of values (v_1, v_2), each occuring with probability 1/4:
$(1,1), (1,2), (2,1)$ and $(2,2)$.

The revenue of the seller is 1 in the first three cases and 2 in the fourth case, so their expected revenue is

$$1 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \frac{5}{4}$$

(b) Now let's suppose that there are three bidders who have independent, private values v_i $(i=1,2,3)$, which are either 1 or 2. As before, the probabilities of values 1 and 2 are 1/2 for every bidder, and if there is a tie at a bid of x for the highest bid, then the winner is selected at random from among the highest bidders and the price is x.

What is the seller's expected revenue in this case?

Now, there are eight possible combinations of values (v_1, v_2, v_3), each occuring with probability $1/8$: $(1,1,1)$, $(1,1,2)$, $(1,2,1)$, $(2,1,1)$, $(1,2,2)$, $(2,1,2)$, $(2,2,1)$ and $(2,2,2)$. The revenue of the seller is 1 in the first four cases and 2 in the last four cases, so their expected revenue is $1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2}$.

(c) Explain why changing the number of bidders affects the seller's expected revenue.

Parts (a) and (b) suggest a conjecture that as the number of bidders increases, the seller's expected revenue also increases. Intuitively, this occures because as the number of bidders grow, the expectation of the second highest bid also grows.

(d) Now let's suppose that the seller sets a reserve price of R with $1 < R < 2$: that is, the object is sold to the highest bidder if his or her bid is at least R, and the price this bidder pays is the maximum of the second highest bid and R. If no bid is at least R, then the object is not sold, and the seller receives 0 revenue.

Suppose that there are two bidders in the auction, who both know R. What is the seller's expected revenue as a function of R?

As in part (a), there are four possible pairs of values (v_1, v_2), each occuring with probability 1/4: $(1,1), (1,2), (2,1), (2,2)$. The revenue of the seller is 0 in the first case, R in the next

two cases and 2 in the last case, so the expected revenue is $R \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = \frac{1}{2}(R+1)$.

(e) Using the previous part, show that a seller who wants to maximise their expected revenue would never set a reserve price, R, that is more than 1 and less than 1.5.

If we compare the seller's expected revenue with reserve price R to their expected revenue without reserve price, we get that the former is greater than or equal to the latter only for R greater than or equal to $\frac{3}{2}$:

$$\frac{1}{2}(R+1) \geq \frac{5}{4}$$

$$R+1 \geq \frac{5}{2}$$

$$R \geq \frac{3}{2}$$

(f) Suppose that the seller now decides to charge an entry fee of 0.5, and there are two potential bidders.

Any bidder who wants to participate in the auction must pay this fee to the seller before bidding and, in fact, this fee is imposed even before each bidder even knows his or her own value for the object. The bidders know only the distribution of values and that anyone who pays the fee will be allowed to participate in a second-price auction for the object. This adds a new first stage to the game in which bidders decide simultaneously whether to pay the fee and enter the auction, or not to pay the fee and stay out of the auction.

This first stage is then followed by a second stage in which anyone who pays the fee participates in the auction. We will assume that after the first stage is over both potential bidders learn their own value for the object (but not the other potential bidder's value for the object) and that they both learn whether or not the other potential bidder decided to enter the auction.

Any potential bidder who does not participate in the auction has a profit of 0. If no one chooses to participate then the seller keeps the object and does not run an auction; if only one bidder chooses to participate in the auction then the seller runs a second-price auction with only this one bidder

(and treats the second highest bid as 0); and finally, if both bidders participate, the second-price auction as in part (a) is run (but note the reduction of the entry fee).

Is there an equilibrium in which each bidder pays the fee and participates in the auction? Give an explanation for your answer.

Let's construct the payoff matrix of the game the bidders play in the first stage. Each bidder has two strategies: participate (Yes) or not participate (No) in the auction. Whenever a bidder chooses to not enter the auction, their payoff is 0, so we only need to calculate their expected utilities for entering the auction.

There're two possible cases: either the bidder is the only participant, or there are two participants in the auction. In the first case, the bidder gets the object for the price of 0, so their expected utility equals their expected value for the object, minus the entry fee:

$$\left(1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2}\right) - \frac{1}{2} = \frac{3}{2} - \frac{1}{2} = 1.$$

In the second case, as in part (a), there are four possible pairs of values (v_1, v_2), each occuring with probability 1/4: (1,1), (1,2), (2,1), (2,2). In the first and last cases where a tie occurs, with probability 1/2 the bidder gets the object but pays as much as they bid, and with probability 1/2 they do not get the object, so their utility from the auction

is 0. In cases without a tie, the bidder gets the object only in the case where they placed a higher bid, in which case their utility from the auction is $2-1=1$, while in the other case they don't get the object and get 0 utility. Hence, the expected utility of the bidder from the auction is $0 \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{1}{4}$. Finally, their expected payoff in the entire game is $\frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$ (minus the entree fee).

We thus get the following payoff matrix:

Bidder 2

Bidder 1	Yes	No
Yes	$-\frac{1}{4}, -\frac{1}{4}$	1, 0
No	0, 1	0, 0

We thus see that the state in which both bidders pay the fee and participate in the auction is not an equilibrium, as well as the state where none of them does. The equilibria of this game are two states in which one of the bidders participate in the auction and the other stays out.

(2) Roughly speaking, the revenue equivalence theorem states that, no matter what the payment rules are, as long as the allocation of items remains the same, in equilibrium, the revenue from all auctions is the same. This means that for a seller, it does not matter what the payments are (in theory).

Discuss why, despite the revenue equivalence theorem, truthful auctions are preferred.

The revenue equivalence theorem makes several assumptions, including the fact that players are rational and play Bayes-Nash equilibrium, where there is a common knowledge of the distribution of their valuations. All these assumptions are not necessary when considering dominant-strategy truthful auctions. For a theory to be useful in practical applications it is better to have fewer assumptions:
Moreover, when true agents' valuations are revealed to the mechanism, an efficient

allocation maximising the social welfare can be implemented.

(3) A mechanism is <u>direct</u> if bidders directly report their private valuation (their "type") to the mechanism, which then calculates the allocation and payments based on these reports (e.g., a sealed-bid Vickrey auction is considered a direct mechanism). Formally, for each agent i, the set of possible messages M_i equals the set of possible types Θ_i, and the mechanism asks to report a type $\hat{\theta}_i \in \Theta_i$. The outcome function then maps the reported type profile $(\hat{\theta}_1, \dots, \hat{\theta}_n)$ to an outcome (allocation + payments).

Any mechanism that is not direct is <u>indirect</u>. These are protocols where bidders interact by acting on information (like price updates) rather than simply reporting their valuation. Examples include the English (ascending) and Dutch (descending) auctions where agents' strategies are dynamic bidding rules rather than one-shot reports of their private values.

Revelation principle

Any social choice function that is implementable in (Bayesian Nash) equilibrium by some mechanism is also implementable by a direct, incentive-compatible mechanism in which agents

optimally reveal their true types. Formally, let

$$f : \Theta_1 \times \ldots \times \Theta_n \rightarrow O$$

be a social-choice function and suppose there exists some mechanism (possibly indirect) and Bayesian Nash equilibrium of that mechanism that implements f (i.e., for every type profile $\theta = (\theta_1, \ldots, \theta_n)$, the equilibrium outcome is $f(\theta)$). Then there exists a direct revelation mechanism in which

- each agent i simply reports a type $\hat{\theta}_i \in \Theta_i$
- truthful reporting $\hat{\theta}_i = \theta_i$ is a Bayesian Nash equilibrium,
- and the outcome of the mechanism on reports $\hat{\theta}$ is exactly $f(\hat{\theta})$.

In other words, anything implementable by some equilibrium of some mechanism is also implementable by a truthful equilibrium of a direct revelation mechanism. This dramatically simplifies analysis: to study what is implementable, we can focus on incentive-compatible direct mechanisms rather than all possible games. The revelation principle is a foundational result in mechanism design that lets you restrict attention to truthful direct mechanisms without loss of generality.

(4) Suppose you are running a VCG-based simultaneous multi-item auction, where three related items A, B, C are being auctioned, and each bidder can bid on any possible subset of items.

Suppose there are two bidders, X and Y, and they provide you with their claimed valuation as their bids for every subset of the items, as part of the bidding process. Suppose that the valuation functions v_X and v_Y that you receive from the two bidders, X and Y, respectively, are as follows:

	$\emptyset$	$\{A\}$	$\{B\}$	$\{C\}$	$\{A,B\}$	$\{A,C\}$	$\{B,C\}$	$\{A,B,C\}$
v_X	0	24	4	9	29	38	20	50
v_Y	0	15	18	11	30	34	32	47

What is the outcome of this VCG auction? Which of the two bidders will get which of the items, and what price will they each pay?

Assume the bidders bid their true valuations (which we can, since in the VCG mechanism

bidding truthfully is a dominant strategy for all bidders).

Then, given these valuations, the VCG mechanism firstly picks an outcome that maximises the total sum of valuations for all the bidders, i.e., maximises the total social welfare of the outcome. In this case, interestingly, there are two different outcomes that maximise the social welfare:

If bidder X gets allocated items {A, C} and player Y gets allocated {B}, then the total valuation of this outcome is 38+18=56. Likewise, if X gets {A} and Y gets {B, C}, then the total valuation is 24+32 = 56.

Furthermore, one can check (exhaustively) that these two outcomes provide the maximum total valuation of any outcome. Therefore, the VCG mechanism picks one of these possible outcomes, and then asks the bidders to pay the VCG payments associated with that outcome.

Note that the VCG mechanism, as given, does not specify which of these outcomes is to be preferred. Either will do! However, these two outcomes are very different and also yield very different payments.

Consider the outcome that gives X the set $\{A, C\}$ and gives Y the set $\{B\}$. We now calculate the payments of X and Y for this outcome. Note that, since

Y is the only player other than X, the VCG transfer for X in this outcome is:

$$t_X = v_Y(\{B\}) - \max_{O \in \text{outcomes}} v_Y(O) =$$

$$= v_Y(\{B\}) - v_Y(\{A, B, C\})$$

$$= 18 - 47 = -29.$$

Similarly, the transfer for Y is:

$$t_Y = v_X(\{A, C\}) - \max_{O \in \text{outcomes}} v_X(O)$$

$$= v_X(\{A, C\}) - v_X(\{A, B, C\})$$

$$= 38 - 50 = -12.$$

That is, bidder X will pay 29 to the mechanism, and bidder Y will pay 12.

On the other hand, if the VCG mechanism chooses the other social-welfare-maximising outcome, namely X gets $\{A\}$ and Y gets $\{B, C\}$, then the payments are as follows:

the transfer for X is:

$$t_X = v_Y(\{B, C\}) - v_Y(\{A, B, C\})$$
$$= 32 - 47 = -15,$$

the transfer for Y is:

$$t_Y = v_X(\{A\}) - v_X(\{A, B, C\})$$
$$= 24 - 50 = -26.$$

That is, bidder X will pay 15 to the mechanism, and bidder Y will pay 26.
In both cases, the mechanism will collect 41 of surplus.